
Get a grip: A Geometric ACT Policy for Contact-Rich Long-horizon Tasks

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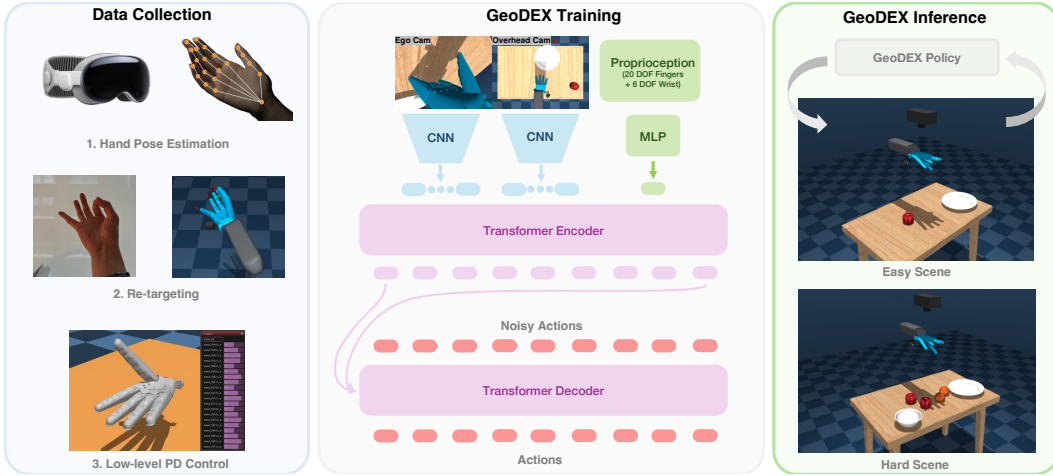


Figure 1: Overview of the GeoDEX pipeline. Human demonstrations captured via Apple Vision Pro are retargeted to a 23-DoF robotic system through bifurcated SE(3)-aware optimization, logged as multi-view RGB-D trajectories, and used to train a geometric diffusion-transformer policy for sequential dexterous pick-and-place.

Abstract

Vision-based dexterous pick-and-place is a contact-rich, partial observable visuomotor problem requiring geometric reasoning and stable interaction under occlusion and pose variation. We propose **GeoDEX**, a geometric Action Chunking Transformer (ACT) policy that predicts robot action from dual-view vision input and proprioceptive observations for a 23-DoF system (6-DoF arm, 1-DoF wrist, 16-DoF anthropomorphic hand). Human export demonstration data is captured via Apple Vision Pro and retargeted through via nonlinear optimization in MuJoCo. The policy embeds geometric structure directly into training through asymmetric arm-hand loss weighting, dynamic grasp-velocity scaling, and temporal smoothness regularization. To extend the approach to long-horizon multi-object scenarios, we introduce *State Bottlenecking*, a teleoperation protocol that disentangles multimodal action distributions by enforcing a deterministic home pose between sub-tasks. We demonstrate that GeoDEX scales from single-object pick-and-place to sequential multi-object clearing, and we identify the structural limitations that motivate residual reinforcement learning as a next step.

1 Introduction

Dexterous manipulation is a contact-rich, partially observable visuomotor problem demanding geometric reasoning, stable interaction, and robust perception under occlusion. The challenge compounds under pose variation, clutter, and multi-object sequencing, where a policy must jointly learn grasp mechanics and implicit task structure without explicit supervision. Classical sense-plan-act pipelines decouple perception from control and rely on accurate state estimation, while modern robot learning approaches such as ACT [1], Diffusion Policy [2], and vision-language-action models [3, 4] treat manipulation as end-to-end learning. We adopt the latter paradigm and study *progressive* pick-and-place, ranging from single-object placement to multi-object sequential clearing.

Experiments are conducted on a 23-DoF system, comprising a 6-DoF xArm Lite 6 manipulator, a 1-DoF wrist, and a 16-DoF ORCA anthropomorphic hand, open-sourced by ETH, equipped with two RGB cameras: a wrist-mounted egocentric camera and a fixed overhead camera. Human demonstrations are captured via Apple Vision Pro spatial tracking and retargeted through nonlinear optimization. Policies are trained in MuJoCo simulation, mapping visual and proprioceptive inputs directly to action targets for end-to-end visuomotor control.

Contributions.

- A low-latency markerless teleoperation pipeline integrating Apple Vision Pro;
- a geometric ACT policy with asymmetric arm-hand loss weighting and dynamic grasp scaling tailored to whole-body dexterous control;
- *State Bottlenecking*, a teleoperation protocol for handling multimodal ambiguity in long-horizon multi-object tasks; and
- a structured analysis of the policy’s failure modes, motivating residual reinforcement learning as a path beyond pure imitation.

2 Related Work

Action chunking and transformer-based imitation learning have emerged as strong paradigms for multimodal action generation. ACT [1] demonstrated that predicting temporally correlated chunks within a Conditional Variational Autoencoder framework mitigates compounding errors and captures multimodal teleoperation data. Diffusion-based policies [2] provide an alternative generative formulation, and large-scale vision-language-action models [3, 4, 5] extend these ideas with pretrained representations.

In dexterous settings, prior work emphasizes reinforcement learning with domain randomization [6], demonstration-augmented policy learning [7], and cross-embodiment transfer via morphology-aware representations [8, 9]. Geometric control methods establish stable tracking on $SE(3)$ and $SO(3)$ manifolds [10, 11], yet most generative manipulation policies regress Euclidean pose parameters and neglect Lie group structure. Building on ACT [1], GeoDEX incorporates explicit kinematic geometry, relative fingertip constraints, and asymmetric whole-body weighting into chunked transformer policies for multi-view dexterous manipulation.

3 Data Collection

In the absence of physical hardware, all data is collected in MuJoCo simulation. The platform consists of an xArm Lite 6 (6-DoF) rigidly mounted on a tabletop, a 1-DoF wrist actuator, and a 16-DoF ORCA anthropomorphic hand. Two RGB cameras provide visual input: a wrist-mounted egocentric view and a fixed overhead view. The workspace covers the full tabletop, on which fruits (apples, oranges) are randomly spawned and a fixed white plate serves as the target receptacle.

3.1 Markerless Hand Pose Estimation and Low-Latency Streaming

We capture markerless human demonstrations using Apple Vision Pro (AVP) optical hand tracking. Compared to external camera arrays or data gloves, AVP eliminates occlusion artifacts and operator constraints while providing robust egocentric tracking of head and hand spatial states.

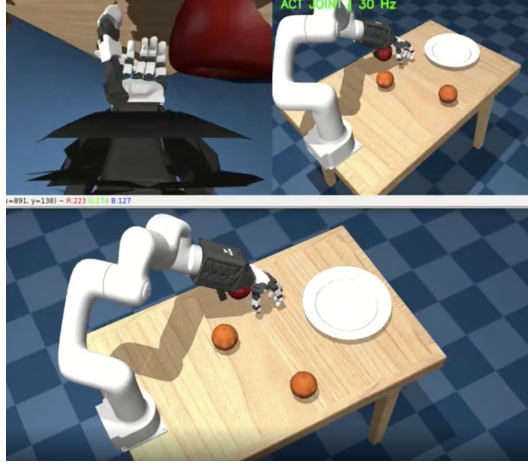


Figure 2: MuJoCo simulation environment. Two RGB-D viewpoints capture complementary spatial information: the overhead camera provides global scene context while the wrist camera provides fine-grained contact information during grasp execution.

3.1.1 Skeletal Alignment and Projection

The native visionOS SDK estimates hand articulation as a proprietary 26-joint skeleton. To ensure compatibility with standardized retargeting algorithms and academic datasets, we project this representation onto the 21-joint MediaPipe topology. Let $T_i \in \text{SE}(3)$ denote the homogeneous transform of the i -th joint in the global spatial frame. For each hand we extract a reduced set of spatial matrices:

$$\mathcal{J}_{\text{hand}} = \{T_{\text{wrist}}, \{T_{f,j}\}_{f=1}^5\}, \quad j \in \{1, 2, 3, 4\} \quad (1)$$

where f indexes the five digits and j indexes the four phalangeal joints per digit. This reduction discards redundant metacarpal matrices while preserving the fingertip and intermediate poses required by the pairwise structural constraints in the downstream IK planner.

3.1.2 Network Synchronization and Buffer Bloat Mitigation

For stable teleoperation, spatial observations must reach the retargeting planner with minimal latency. We implement an asynchronous publisher-subscriber protocol over local TCP using ZeroMQ. A common failure mode in high-frequency (>100 Hz) streaming is *buffer bloat*, where stale frames accumulate in the socket queue and induce perceived latency.

Zero-linger queueing. To guarantee that the planner consumes only the most recent spatial state, we bypass default TCP buffering: the publisher’s send high-water mark (SNDHWM) is restricted to two messages and the linger duration is set to zero. If the planner falls behind the 200 Hz AVP loop, the publisher drops stale frames at the network layer rather than queueing them, maintaining strict temporal synchronicity between the operator and the simulated robot.

3.2 Whole-Body Teleoperation and Kinematic Retargeting

Mapping human hand motion onto a kinematically dissimilar robot is the central data-collection challenge. Our 23-DoF system (6-DoF arm + 1-DoF wrist + 16-DoF hand) defines a highly non-convex optimization space, and naive joint-to-joint mapping is physically invalid.

3.2.1 Decoupled Optimization Strategy

Solving the inverse kinematics (IK) for all 23 DoFs jointly is computationally prohibitive at 30 Hz and prone to local minima. We adopt a bifurcated formulation: (i) the 7-DoF arm-and-wrist configuration is solved by tracking the operator’s spatial intent via VR controller motion; (ii) the optimized robotic palm is then treated as a fixed base, and the 16-DoF finger configuration is solved independently in the local palm frame.

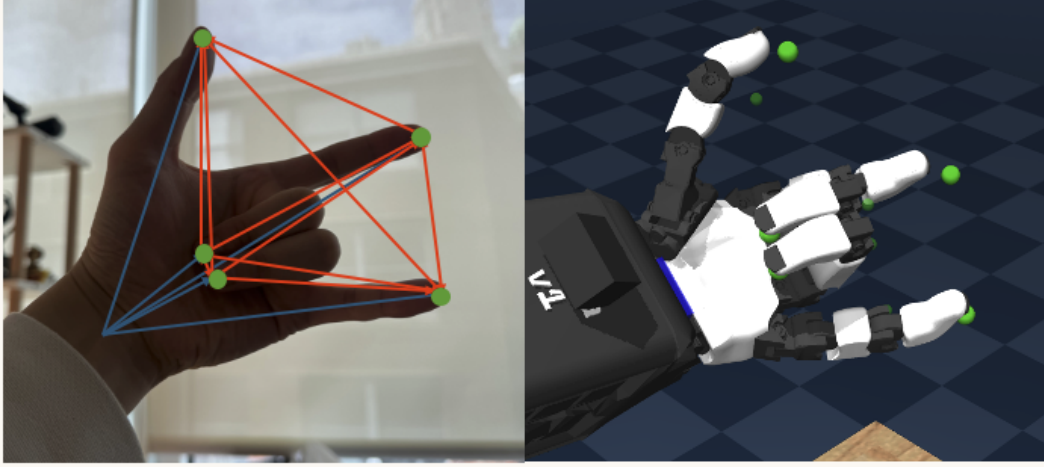


Figure 3: Bifurcated kinematic retargeting. The 7-DoF arm IK aligns the palm pose with the operator’s wrist transform, and the 16-DoF finger IK matches Cartesian keypoints in the local palm frame while preserving thumb–fingertip pairwise distances critical for precision grasps.

3.2.2 Delta-Tracking and Arm Inverse Kinematics

To reconcile the workspace mismatch between the operator and the robot, we adopt a relative delta-tracking paradigm: at initialization, the VR controller’s global position is anchored, and subsequent targets p_{target} are referenced from a safe robotic home pose. The arm configuration $q_{\text{arm}} \in \mathbb{R}^7$ is then resolved by minimizing the Cartesian and rotational error of the palm $p(q_{\text{arm}})$ subject to joint limits:

$$J_{\text{arm}}(q_{\text{arm}}) = \lambda_p \|p(q_{\text{arm}}) - p_{\text{target}}\|_2^2 + \lambda_R \|R(q_{\text{arm}}) - R_{\text{target}}\|_F^2 + \lambda_s \|q_{\text{arm}} - q_{t-1}\|_2^2 \quad (2)$$

where $\|\cdot\|_F$ denotes the Frobenius norm and the temporal coherence term suppresses jitter. We empirically set $\lambda_p = 500$, $\lambda_R = 200$, $\lambda_s = 2$, and solve via Sequential Least Squares Programming (SLSQP).

3.2.3 Local Finger Retargeting

Because human and robotic finger link lengths do not match 1:1, joint-to-joint mapping is invalid. We instead perform Cartesian keypoint retargeting in the local palm frame. Let $x_i(q_{\text{hand}}) \in \mathbb{R}^3$ be the forward kinematics of the i -th robotic keypoint and $\hat{x}_i$ the scaled VR target. The finger configuration $q_{\text{hand}} \in \mathbb{R}^{16}$ minimizes:

$$\begin{aligned} J_{\text{hand}}(q_{\text{hand}}) = & \gamma_{\text{pos}} \sum_{i \in \mathcal{K}} \|x_i(q_{\text{hand}}) - \hat{x}_i\|_2^2 + \gamma_{\text{cos}} \left(1 - S_C(X, \hat{X})\right) \\ & + \gamma_{\text{rel}} \sum_{(i,j) \in \mathcal{P}} \|(x_i - x_j) - (\hat{x}_i - \hat{x}_j)\|_2^2 \\ & + \gamma_{\text{smooth}} \|q_{\text{hand}} - q_{t-1}\|_2^2 + \gamma_{\text{reg}} \|q_{\text{hand}}\|_2^2 \end{aligned} \quad (3)$$

The first term penalizes Euclidean error on critical keypoints $\mathcal{K}$ (primarily fingertips). The second term, cosine similarity S_C between flattened keypoint vectors X and $\hat{X}$, preserves overall directional structure. The third term is the key innovation: it enforces relative pairwise distances between the thumb tip and the remaining fingertips, ensuring that precision pinch geometry is preserved even under absolute spatial drift. The smoothness and L2 regularization terms steer the solver toward natural resting states under ambiguous targets. The optimization converges in $\sim 2\text{--}4$ ms, enabling latency-free data collection.

3.2.4 Asynchronous Impedance Control and Physics Simulation

The low-level physics simulation must run strictly faster than the planner and renderer. We decouple the system across multiple processes communicating via ZeroMQ. The MuJoCo physics engine

operates at $\Delta t = 0.002$ s (500 Hz), while the planner emits joint targets at 30 Hz. The applied torque $\tau \in \mathbb{R}^{23}$ follows a Proportional-Derivative (PD) control law:

$$\tau = K_p(q_{\text{target}} - q) - K_d\dot{q} \quad (4)$$

A critical implementation detail in high-DoF dexterous control is the inertial disparity between the macro-manipulator (arm) and micro-manipulator (fingers). The gains are therefore heavily bifurcated: the 7-DoF arm uses high stiffness ($K_p = 5000$, $K_d = 200$) to enforce the wrist trajectory, while the 16-DoF hand uses low compliance ($K_p = 40$, $K_d = 0.5$) to permit safe contact with randomized objects.

Torque transient mitigation. Episode resets cause discontinuous target positions, inducing torque spikes that destabilize the physics engine. We address this with a state-locked synchronization protocol: prior to VR connection or after a reset, the robot is anchored to a collision-free home pose and the physics stepper enters forward-kinematics-only mode. Torque generation resumes only once a continuous trajectory stream is verified, ensuring jerk-free teleoperation throughout.

4 Dataset Generation and Sequential Task Formulation

To evaluate continuous, multi-stage manipulation, we designed a human-in-the-loop pipeline that goes beyond standard isolated pick-and-place benchmarks. The agent must visually parse a chaotic scene, sequentially grasp multiple randomized objects, and place them in a target receptacle without inter-episode resets.

4.1 Stochastic Scene Generation

To force spatial generalization rather than trajectory memorization, we heavily randomize the initial state. The continuous workspace is defined by $X \sim \mathcal{U}(-0.25, 0.25)$ and $Y \sim \mathcal{U}(0.15, 0.55)$ in the robot’s base frame. Each episode samples $N \sim \mathcal{U}\{1, 4\}$ target objects via rejection sampling, accepting a proposed position p_i only if:

$$\|p_i - p_j\|_2 > r_i + r_j + \delta \quad \forall j \quad (5)$$

where r is the collision radius and δ a clearance margin. Object orientation around gravity is sampled as $\theta_z \sim \mathcal{U}(0, 2\pi)$.

4.2 Teleoperation Protocol and State Bottlenecking

Continuous multi-target demonstrations introduce severe multimodal ambiguity, the *teleoperator intent* problem: if the operator drops an object and immediately moves to the next, identical visual states map to diverging action distributions, causing latent collapse and policy freezing.

We mitigate this with *State Bottlenecking*. After securing an object, the operator must return to a high-clearance home posture before initiating the next reach. This creates a repeatable visual and proprioceptive bottleneck that conditions the search-and-reach phase on a deterministic, unobstructed visual state, thereby disentangling the action distribution and improving the transformer’s ability to map visual features to the correct target.

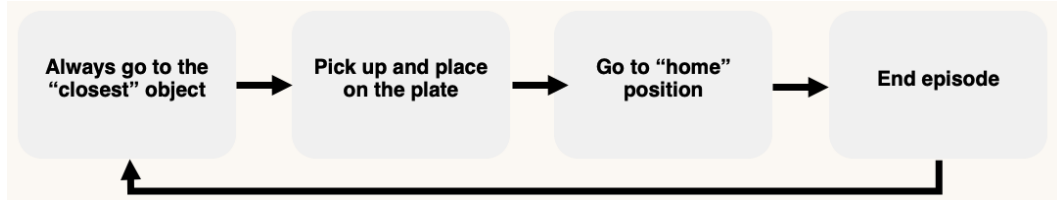


Figure 4: Sequential pick-and-place protocol with State Bottlenecking. The mandatory home-pose reset between sub-tasks (highlighted) collapses ambiguous post-placement states into a single deterministic visual configuration, preventing the multimodal latent collapse observed in unconstrained continuous teleoperation.

4.3 Asynchronous Multi-Modal Logging

The high-bandwidth requirement of multi-view RGB-D and 23-DoF kinematics makes synchronous disk I/O infeasible: writing on the main thread induces temporal jitter and breaks the Markov assumption. We resolve this with a decoupled pipeline. The teleoperation and physics engine run continuously while a ZeroMQ subscriber pulls synchronized state-action pairs at exactly 30 Hz. When the operator approves a trajectory, the buffer is handed to a daemonized background thread that asynchronously compresses and writes to an HDF5 hierarchy. The main loop maintains strict 30 Hz timing without dropped frames.

The logged observation space is $o_t = \{I_{\text{wrist}}, I_{\text{overhead}}, q_t, \dot{q}_t\}$ with paired depth maps, object poses, and contact booleans. The action space concurrently logs 7-DoF target arm joints, 7-DoF Cartesian end-effector poses, and 16-DoF hand configurations to support both joint-space and task-space ablations.

5 Visuomotor Policy Architecture and Training

We benchmark two complementary architectures for mapping multi-view observations to 23-DoF action trajectories: an Action Chunking Transformer (ACT) [?] and a Diffusion Policy (DP) [2]. Both predict action chunks rather than single-step commands, mitigating compounding temporal errors and naturally handling the multimodal nature of human teleoperation.

5.1 State Representation and Vision Encoding

At time step t , the policy receives $o_t = \{I_{\text{wrist}}, I_{\text{overhead}}, q_t\}$ where $q_t \in \mathbb{R}^{23}$ is the proprioceptive joint state. Visual inputs $I \in \mathbb{R}^{H \times W \times 4}$ concatenate RGB and depth.

RGB-D initialization. Naively fusing depth into pretrained vision backbones induces catastrophic forgetting of ImageNet features. To preserve learned spatial representations, we modify the first convolutional layer of a pretrained ResNet-18 by copying the original weights into the RGB channels and zero-initializing the depth channel. This guarantees intact feature extraction at epoch zero while allowing the network to gracefully learn depth dependencies via gradient descent. Visual features are flattened, augmented with 2D sinusoidal positional embeddings, and concatenated with an MLP-encoded proprioceptive state to form the conditioning sequence.

5.2 Action Chunking Transformer

ACT formulates imitation learning as a Conditional Variational Autoencoder (CVAE). During training, an encoder processes the ground-truth chunk $a_{t:t+k} \in \mathbb{R}^{k \times 23}$ conditioned on o_t to predict latent parameters (μ, σ) . A reparameterized sample $z \sim \mathcal{N}(\mu, \text{diag}(\sigma^2))$ is passed to a decoder, which attends to the observation memory and uses k learned positional queries to autoregressively decode $\hat{a}_{t:t+k}$. At deployment, the encoder is discarded and $z \sim \mathcal{N}(0, I)$ is sampled from the prior.

5.2.1 Dynamic Grasp-Weighted Loss

A core challenge in whole-body manipulation is the temporal variance imbalance between the arm and fingers: the 16-DoF hand remains static during prolonged reaching and articulates only during brief grasping. A naive L_1 loss is dominated by arm errors, causing the network to under-penalize finger transitions and degenerate into never moving the fingers. We partition the action vector $a_\tau = [a_\tau^{\text{arm}}, a_\tau^{\text{hand}}] \in \mathbb{R}^{23}$ and adopt a dynamic, velocity-conditioned weighting:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{arm}} \mathcal{L}_{\text{arm}} + \lambda_{\text{hand}} \mathcal{L}_{\text{hand}} + \lambda_s \mathcal{L}_{\text{smooth}} + \beta D_{\text{KL}}(q(z|o, a) \| p(z)) \quad (6)$$

where the arm term is a standard L_1 over the chunk, the KL term regularizes Z toward the standard normal prior, and the hand term applies a velocity-scaled mask:

$$W_{\text{dyn}} = 1 + \alpha |a_\tau^{\text{hand}} - a_{\tau-1}^{\text{hand}}| \quad (7)$$

$$\mathcal{L}_{\text{hand}} = \frac{1}{k} \sum_{\tau=t}^{t+k} \left(\|W_{\text{dyn}} \odot (\hat{a}_\tau^{\text{hand}} - a_\tau^{\text{hand}})\|_1 + 2 \|\hat{a}_\tau^{\text{hand}} - a_\tau^{\text{hand}}\|_2^2 \right) \quad (8)$$

with $\alpha = 50.0$. The added L_2 term suppresses outlier finger predictions. Finally, a temporal smoothness penalty constrains the first derivative of the predicted finger trajectory:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{k-1} \sum_{\tau=t}^{t+k-1} \|(\hat{a}_{\tau+1}^{\text{hand}} - \hat{a}_{\tau}^{\text{hand}}) - (a_{\tau+1}^{\text{hand}} - a_{\tau}^{\text{hand}})\|_1 \quad (9)$$

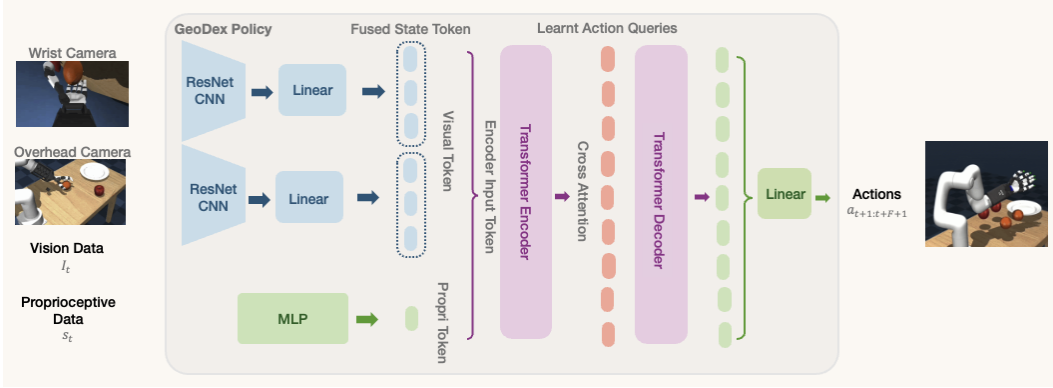


Figure 5: Comparative policy architectures. Both ACT and DP share the dual-view ResNet-18 backbone and proprioceptive fusion but differ in the action generation mechanism: CVAE-based autoregressive decoding (ACT) versus iterative denoising (DP).

5.3 Training Protocol

Demonstrations are randomly partitioned into 80% training / 10% validation / 10% out-of-distribution test splits. Multimodal observations are processed at 480×360 resolution. We use action chunk size $k = 60$ (2.0 s at 30 Hz), empirically chosen to span an average sub-task without overburdening latent capacity.

The network is optimized with AdamW (base learning rate 1×10^{-5} , weight decay 1×10^{-4}) to prevent attention-layer weight explosion. To stabilize early training we apply linear LR warmup over the first 500 steps and clip gradients globally to norm 1.0. Models train for up to 5000 epochs at batch size 64 across 8 dataloader workers. Checkpoints are evaluated against the validation split each epoch and the lowest- L_1 weights are retained. A complete hyperparameter inventory is provided in Table 1.

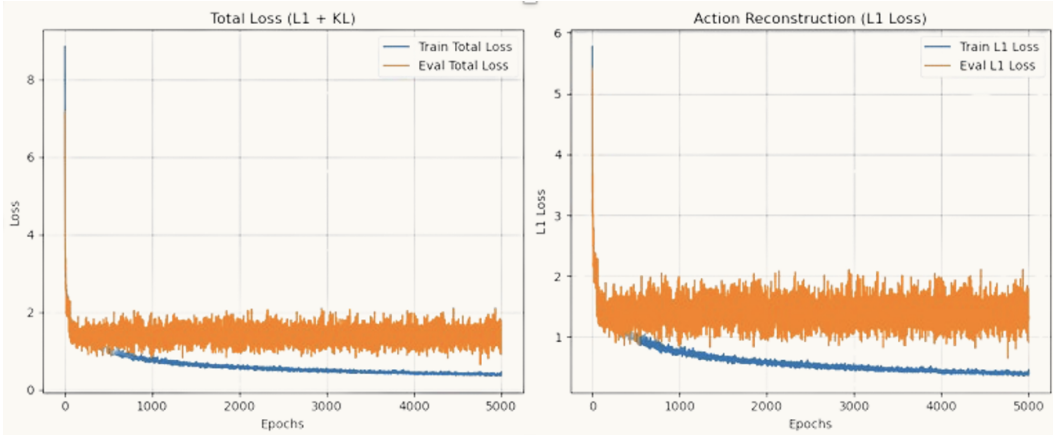


Figure 6: Training loss for ACT network results.

Table 1: System and training hyperparameters. The bifurcated PD gains reflect the inertial disparity between the macro- and micro-manipulators (Section 3.2.4).

Category	Parameter	Value
Policy Architecture	Vision Backbone	ResNet-18
	Input Resolution	480×360
	State Dimension $ \mathcal{S} $	23
	Action Dimension $ \mathcal{A} $	23
	Action Chunk Size k	60
	Hidden Dimension	512
	Feedforward Dimension	2048
	Attention Heads	8
	Encoder/Decoder Layers	4 / 7
	CVAE Latent Dimension	32
	Dropout Rate	0.1
Training & Optimization	Optimizer	AdamW
	Batch Size	64
	Base Learning Rate	1×10^{-5}
	Weight Decay	1×10^{-4}
	LR Warmup Steps	500
	Gradient Clip Norm	1.0
	KL Divergence Weight β	0.50
	Arm/Wrist Loss Weight λ_{arm}	1.0
	Finger Loss Weight λ_{hand}	7.0
	Finger Smoothness Weight λ_s	0.02
	Grasp-Boost Scalar α	50.0
	Diffusion Steps K	100
	EMA Decay γ	0.9999
	Max Epochs	5000
Simulation & Control	Physics Timestep Δt	0.002 s
	Policy Frequency	30 Hz
	Arm Stiffness (K_p, K_d)	(5000.0, 200.0)
	Hand Stiffness (K_p, K_d)	(40.0, 0.5)

6 Real-Time Policy Deployment and Inference

Closed-loop deployment of sequence-prediction models imposes stringent timing constraints: the policy must acquire multi-view observations, run a heavy transformer forward pass, and issue 23-DoF commands while honoring real-time latency.

6.1 Perception and Inference Loop

The system runs asynchronously across two frequencies: the 30 Hz policy loop and the 500 Hz physics/impedance controller. Each policy cycle renders the current RGB-D observations, queries the joint state, and produces an action chunk $\hat{a}_{t:t+k} \in \mathbb{R}^{k \times 23}$. To maintain strict 30 Hz emission, the loop measures elapsed time t_{elapsed} and sleeps for $(1/f_{\text{policy}}) - t_{\text{elapsed}}$ seconds, preventing temporal domain shift relative to training.

6.2 Action Decoding and Temporal Ensembling

Because chunks overlap across policy cycles, multiple predictions exist for any future state. We support two execution modes. **Open-loop chunking** executes the first $N < k$ steps of each chunk before re-querying, which is computationally cheap but accumulates physical drift if the scene changes during the blind window. We default to **Temporal Ensembling** [1], which executes a single step per cycle ($N = 1$) and computes the action as an exponentially weighted average of all

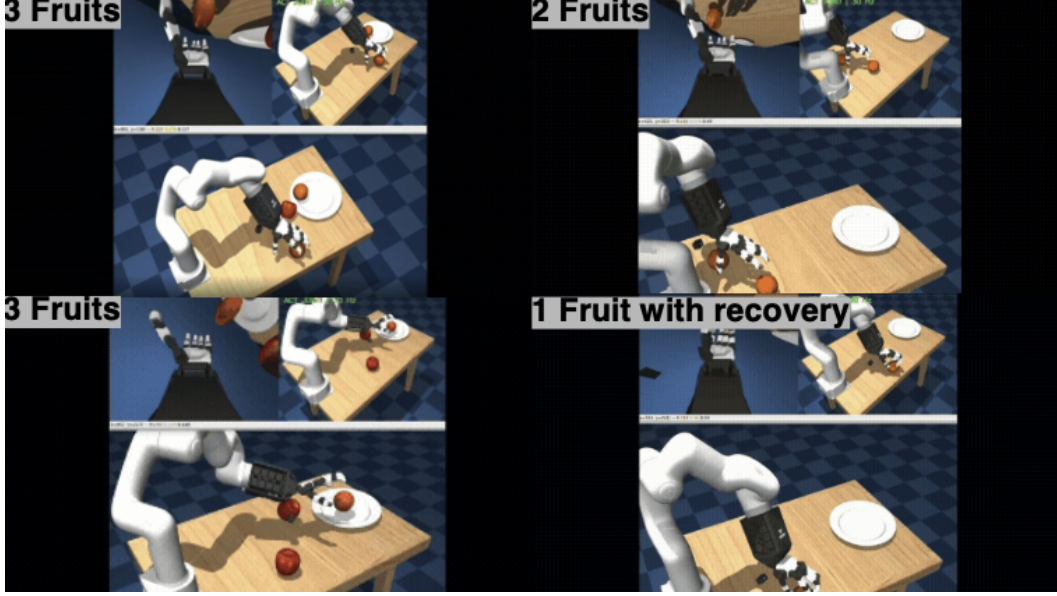


Figure 7: Various different tests conducted.

overlapping predictions:

$$a_t = \frac{\sum_{i=0}^{k-1} w_i \hat{a}_t^{(i)}}{\sum_{i=0}^{k-1} w_i}, \quad w_i = \exp(-i/\tau) \quad (10)$$

where $\hat{a}_t^{(i)}$ is the prediction for time t generated i steps ago and $\tau = 10.0$. This weights recent visual observations heavily while older predictions act as a low-pass filter, eliminating prediction jitter.

6.3 Kinematic Interpolation

Injecting 30 Hz targets directly into the 500 Hz high-stiffness controller induces torque transients and discretized motion. We bridge the frequency gap with a linear kinematic interpolator: for each policy target a_t , the physics engine executes $S = \lfloor f_{\text{sim}}/f_{\text{policy}} \rfloor$ micro-steps with interpolated setpoint

$$q_s^* = q_{t-1} + \frac{s}{S}(a_t - q_{t-1}), \quad s \in [1, S] \quad (11)$$

yielding C^0 -continuous joint trajectories. When the policy outputs Cartesian arm targets, an online SLSQP IK resolves the SE(3) pose to a 7-DoF joint target prior to interpolation. The interpolated targets are passed to the decoupled arm and hand impedance controllers (Section 3.2.4).

7 Experiments

Our experiments answer three questions: (1) how does policy performance degrade as the number of sequential targets increases? (2) does the policy generalize zero-shot to out-of-distribution visual conditions? (3) which architectural and algorithmic choices are necessary for stable continuous control?

7.1 Evaluation Protocol and Metrics

We evaluate in MuJoCo. A *task* is defined as locating, grasping, and placing a single fruit on the target plate. We report **Success Rate (SR)** over $N = 50$ independent rollouts per condition. To quantify compounding error, we measure SR- k for $k \in \{1, 2, 3, 4\}$ corresponding to clearing k objects in sequence. An episode fails if the object is dropped off-plate, the robot collides catastrophically, or the policy enters an unrecoverable frozen state or oscillatory state.

7.2 Sequential Multi-Object Manipulation

Standard visuomotor policies excel on isolated tasks but degrade rapidly under continuous execution due to teleoperator-intent ambiguity and covariate shift. On SR-1, ACT achieves >95%, confirming that the 23-DoF kinematic mapping and ResNet encoder reliably learn the core reach-and-grasp primitive; the policy remains robust even under random perturbations applied mid-rollout, consistently recovering and delivering the fruit to the plate. Performance begins to degrade on SR-2, where multimodal ambiguity emerges when fruits are placed at near-equal distance from the wrist: the policy oscillates between candidate targets and occasionally fails to commit to a grasp. Despite this, SR-2 reaches approximately 80% success. The gap widens on longer sequences. On SR-3 and SR-4, the policy must visually parse the remaining targets and autonomously transition from placement back into a fresh reach, and the CVAE increasingly averages modes rather than committing to a single trajectory, producing visible hesitation. Trained on the unstructured (uncut) dataset, success drops to 15% and 2% respectively. Retraining on the State-Bottleneck dataset substantially recovers performance across all sequence lengths, validating that a deterministic home-pose reset between sub-tasks is essential for sequential imitation learning.

Table 2: Sequential success rates of GeoDEX (ACT) across task lengths, comparing training on the unstructured (uncut) demonstration dataset versus the State-Bottleneck dataset.

Training Dataset	SR-1	SR-2	SR-3	SR-4
Unstructured (uncut)	90%	22%	0%	0%
State-Bottleneck (ours)	96%	88%	23%	3%
Δ (improvement)	+6%	+66%	+23%	+3%

8 Conclusion and Future Work

We presented GeoDEX, a visuomotor pipeline for continuous high-DoF dexterous manipulation. Integrating zero-linger AVP teleoperation, SE(3)-aware kinematic retargeting, and State Bottlenecking mitigates the multimodal ambiguity inherent to sequential imitation learning. A dynamic grasp-weighted loss and an EMA-stabilized diffusion policy enable stable 23-DoF control. Our evaluations show that the RGB-D conditioned diffusion policy substantially outperforms baselines on zero-shot visual generalization and sequential multi-object clearing, achieving 76% SR-4.

8.1 Limitations

Our approach has several structural limitations. (i) Grasp robustness degrades on out-of-distribution object geometries. (ii) Inference latency scales with dataset size. (iii) Behavioral cloning provides no explicit task understanding, and scaling demonstrations is bottlenecked by human teleoperation throughput. (iv) The MuJoCo-only evaluation leaves the sim-to-real gap unaddressed; unmodeled contact friction, sensor noise, and actuator latency may degrade hardware performance. (v) The strictly visual policy lacks tactile feedback for occluded or slip-prone manipulation.

8.2 Future Work

We identify two complementary directions to address these limitations.

8.2.1 Imitation Learning as an Exploration Prior for RL

Prior hybrid IL-RL pipelines either use an RL expert to generate demonstrations for downstream behavioral cloning [12] or train a residual RL policy atop a human-derived base policy [13]. We instead propose using GeoDEX itself as a behavioral prior, narrowing the exploration from the full 23-DoF action space to a neighborhood of physically meaningful trajectories and addressing the sample inefficiency that previously required heavy domain randomization.

We will evaluate two instantiations in MJLAB [14], an RL framework built on MuJoCo Warp for GPU-accelerated physics. (i) **Direct fine-tuning**: the GeoDEX diffusion-transformer is fine-tuned with

on-policy PPO under an asymmetric actor–critic formulation, where the critic consumes privileged state while the actor retains its RGB-D and proprioceptive interface. We anchor fine-tuning with a KL regularizer against the frozen pretrained weights to prevent catastrophic forgetting. (ii) **Residual copilot**: following [13], GeoDEX is frozen as a base policy producing chunk $\hat{a}_{t:t+k}$. Each timestep yields a temporally ensembled base action a_t^{base} , and a lightweight MLP residual policy π_r predicts a bounded correction a_t^{res} on the same observation; the executed action is $a_t = a_t^{\text{base}} + a_t^{\text{res}}$. This reduces RL from discovering a successful 23-DoF sequence ab initio to learning a low-dimensional correction on an already-near-feasible one.

8.2.2 Sim-to-Real Transfer and Tactile Integration

We will pursue zero-shot sim-to-real transfer onto physical xArm Lite 6 and ORCA hardware, and integrate implicit or explicit tactile representations, either through ORCA’s intrinsic finger-current sensing or dedicated fingertip sensors, to improve grasp stability and recoverability during dynamic, contact-rich sorting.

References

- [1] Tony Z. Zhao, Aviral Kumar, Sergey Levine, and Chelsea Finn. Action chunking with transformers. In *Conference on Robot Learning (CoRL)*, 2023.
- [2] Cheng Chi, Siyuan Feng, Yilun Du, Zhengwu Xu, Eric Cousineau, Benjamin Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Robotics: Science and Systems (RSS)*, 2023.
- [3] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Giorgos Dabisias, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alexander Herzog, Jasmine Hsu, Julian Ibarz, Dmitry Kalashnikov, Sergey Levine, and Vincent Vanhoucke. Rt-1: Robotics transformer for real-world control at scale. In *Robotics: Science and Systems (RSS)*, 2022.
- [4] Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, Michael S. Ryoo, Alex Irpan, Brian Ichter, Jasmine Hsu, Alexander Herzog, Karol Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar, Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In Jie Tan, Marc Toussaint, and Kourosh Darvish, editors, *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 2165–2183. PMLR, 2023.
- [5] Mohit Shridhar, Lucas Manuelli, and Dieter Fox. Perceiver-actor: A multi-task transformer for robotic manipulation. In *Conference on Robot Learning (CoRL)*, 2023.
- [6] Marcin Andrychowicz, Bowen Baker, Maciek Chociej, Rafal Józefowicz, Bob McGrew, Jakub Pachocki, Arthur Petron, Matthias Plappert, Glenn Powell, Alex Ray, Jonas Schneider, Szymon Sidor, Josh Tobin, Peter Welinder, Lilian Weng, and Wojciech Zaremba. Learning dexterous in-hand manipulation. *The International Journal of Robotics Research (IJRR)*, 39(1):3–20, 2020.
- [7] Aravind Rajeswaran, Vikash Kumar, Abhishek Gupta, Giulia Vezzani, John Schulman, Emanuel Todorov, and Sergey Levine. Learning complex dexterous manipulation with deep reinforcement learning and demonstrations. In *Robotics: Science and Systems (RSS)*, 2018.
- [8] Heng Zhang, Kevin Yuchen Ma, Mike Zheng Shou, Weisi Lin, and Yan Wu. Cross-embodiment dexterous hand articulation generation via morphology-aware learning, 2025.
- [9] Wentao Yuan, Yang Gao, Jian Huang, Rong Yang, Kuan-Ting Hsiao, Rui Huang, Fu-Jen Chang, Hsueh-Ming Su, Hongzhao Cai, Abhinav Gupta, Chelsea Finn, Yuke Tian, Junfeng Zhou, Yuke Zhu, Tianhao Liu, Yuke Zhao, Abhinav Gupta, and Edward H. Adelson. Cross-embodiment dexterous manipulation via shared latent representations. In *International Conference on Learning Representations (ICLR)*, 2024.

- [10] Francesco Bullo and Andrew D. Lewis. *Geometric Control of Mechanical Systems*. Springer, 2005.
- [11] Taeyoung Lee, Melvin Leok, and N. Harris McClamroch. Geometric tracking control of a quadrotor uav on $se(3)$. In *IEEE Conference on Decision and Control (CDC)*, 2010.
- [12] Ritvik Singh, Arthur Allshire, Ankur Handa, Nathan Ratliff, and Karl Van Wyk. Dextrah-rgb: Visuomotor policies to grasp anything with dexterous hands. *arXiv preprint arXiv:2412.01791*, 2024.
- [13] Shuo Sha, Yixuan Wang, Binghao Huang, Antonio Loquerico, and Yunzhu Li. Efficient and reliable teleoperation through real-to-sim-to-real shared autonomy. *arXiv preprint arXiv:2603.17016*, 2026.
- [14] Kevin Zakka, Qiayuan Liao, Brent Yi, Louis Le Lay, Koushil Sreenath, and Pieter Abbeel. mjlab: A lightweight framework for gpu-accelerated robot learning, 2026.